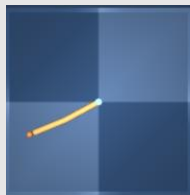


Single-frame JEPA

Single Frame o_t



Encoder



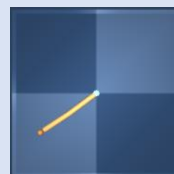
Latent: $z_t = E(o_t)$
(No coordinate for $\dot{q}$)

a_t

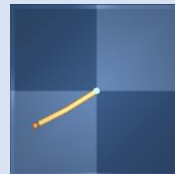
Predictor
(a_t, z_t)

$\hat{z}_{t+1}$

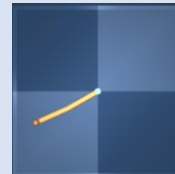
TI-JEPA



O_{t-2}



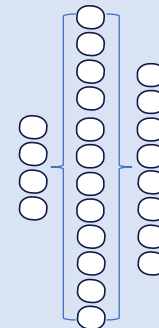
O_{t-1}



O_t

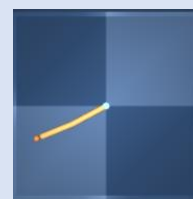
q_Encoder

Δ



Bias-free MLP

v_t



O_t

q_Encoder

SIGReg on q only



Pose q_t

Now I can tell
the Velocity

a_t

memoryless predictor
(q_t, v_t, a_t)

$\hat{q}_{t+1}$

$\hat{v}_{t+1}$